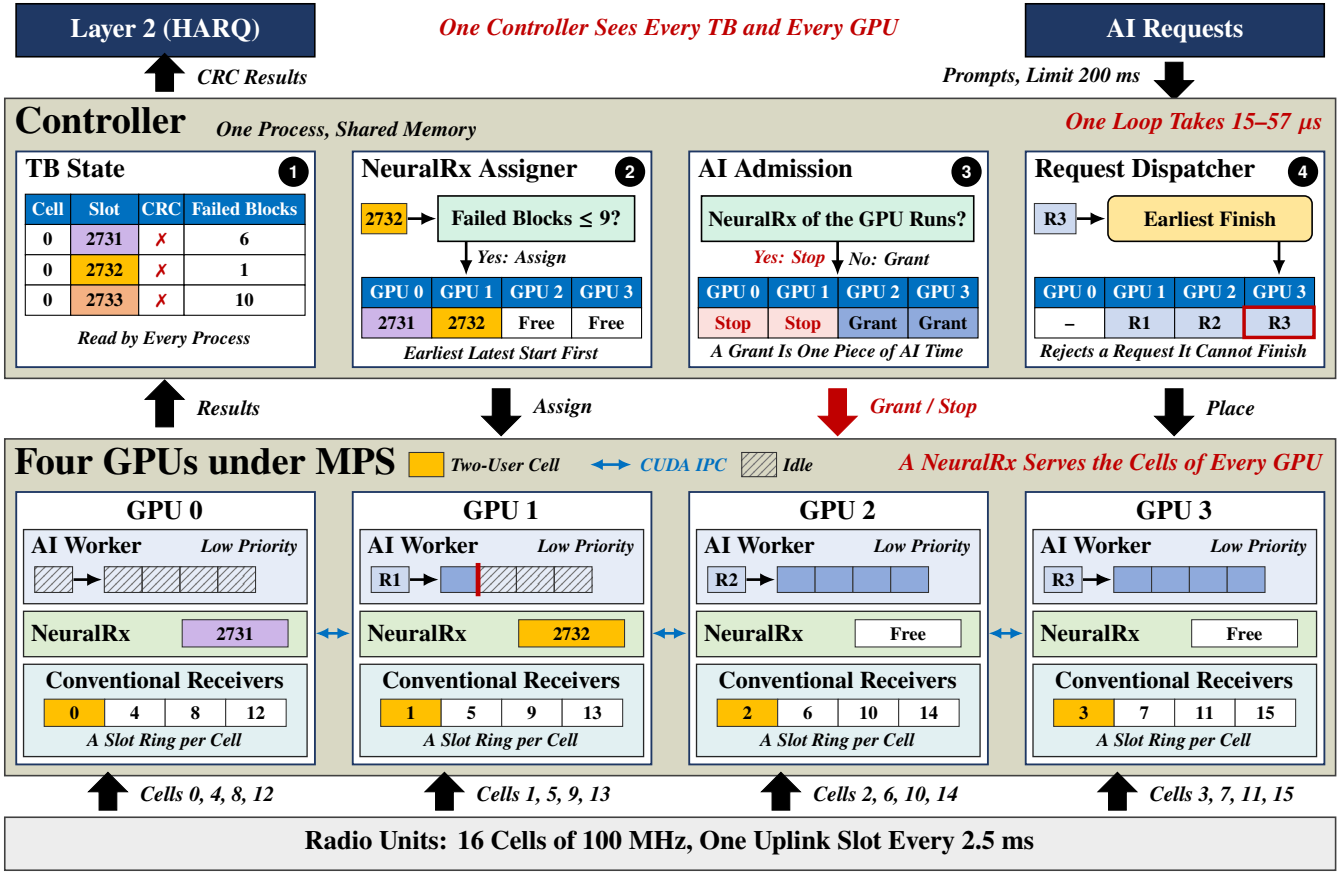

Layer 2 (HARQ)
One Controller Sees Every TB and Every GPU
AI Requests
CRC Results
Prompts, Limit 200 ms
Controller
One Process, Shared Memory
One Loop Takes 15–57 µs
TB State
1
Cell
Slot
CRC
Failed Blocks
0
2731
✗
6
0
2732
✗
1
0
2733
✗
10
Read by Every Process
NeuralRx Assigner
2
2732
Failed Blocks ≤ 9?
Yes: Assign
GPU 0
GPU 1
GPU 2
GPU 3
2731
2732
Free
Free
Earliest Latest Start First
AI Admission
3
NeuralRx of the GPU Runs?
Yes: Stop
No: Grant
GPU 0
GPU 1
GPU 2
GPU 3
Stop
Stop
Grant
Grant
A Grant Is One Piece of AI Time
Request Dispatcher
4
R3
Earliest Finish
GPU 0
GPU 1
GPU 2
GPU 3
–
R1
R2
R3
Rejects a Request It Cannot Finish
Results
Assign
Grant / Stop
Place
Four GPUs under MPS
Two-User Cell
CUDA IPC
Idle
A NeuralRx Serves the Cells of Every GPU
GPU 0
AI Worker
Low Priority
NeuralRx
2731
Conventional Receivers
0
4
8
12
A Slot Ring per Cell
GPU 1
AI Worker
Low Priority
R1
NeuralRx
2732
Conventional Receivers
1
5
9
13
A Slot Ring per Cell
GPU 2
AI Worker
Low Priority
R2
NeuralRx
Free
Conventional Receivers
2
6
10
14
A Slot Ring per Cell
GPU 3
AI Worker
Low Priority
R3
NeuralRx
Free
Conventional Receivers
3
7
11
15
A Slot Ring per Cell
Cells 0, 4, 8, 12
Cells 1, 5, 9, 13
Cells 2, 6, 10, 14
Cells 3, 7, 11, 15
Radio Units: 16 Cells of 100 MHz, One Uplink Slot Every 2.5 ms